

# IT\_4325 INTRODUCTION TO EXTENDED REALITY APPLICATION DEVELOPMENT & DIGITAL TWINS - Remidterm

Points: -/30

Each question has one correct answer. Attempt all 30 questions. 

 **Will be reviewed**

1. 1. Which Unity component determines how a GameObject appears visually? 

- ☐ Collider
- ☒ Mesh Renderer
- ☐ Rigidbody
- ☐ Animator Correct Answer: Mesh Renderer

 **Will be reviewed**

2. 2. In Unity XR, the XR Origin object defines: 

- ☐ Player's start menu
- ☒ The tracking reference frame for the headset
- ☐ The lighting origin
- ☐ The physics center Correct Answer: The tracking reference frame for the headset

 **Will be reviewed**

3. 3. What does the URP (Universal Render Pipeline) primarily improve? 

- ☐ Only performance in VR
- ☒ Customizable, efficient rendering
- ☐ GPU particle simulation
- ☐ CPU-based rendering Correct Answer: Customizable, efficient rendering

 **Will be reviewed**

4. 4. In C#, which keyword ensures a variable retains its value between scene reloads?



- ☐ const
- ☒ static
- ☐ public
- ☐ serialized Correct Answer: static

 **Will be reviewed**

5. 5. The XR Interaction Toolkit allows developers to: 

- ☐ Design 2D UIs
- ☒ Implement controller-based interactions
- ☐ Bake lighting
- ☐ Create physics materials Correct Answer: Implement controller-based interactions

 **Will be reviewed**

6. 6. In Unity physics, which method provides the most stable simulation? 

- ☐ Update()
- ☒ FixedUpdate()
- ☐ LateUpdate()
- ☐ Start() Correct Answer: FixedUpdate()

 **Will be reviewed**

7. 7. What is occlusion culling used for? 

- ☒ Hiding objects behind others to improve performance
- ☐ Reducing texture size
- ☐ Baking lighting
- ☐ Adjusting FOV Correct Answer: Hiding objects behind others to improve performance

 **Will be reviewed**

8. 8. In VR rendering, asynchronous reprojection helps to: 

- ☐ Reduce GPU temperature
- ☒ Maintain visual stability during frame drops
- ☐ Increase FOV
- ☐ Add anti-aliasing Correct Answer: Maintain visual stability during frame drops

 **Will be reviewed**

9. Which component is mandatory for detecting trigger events in Unity? 

- ☒ Collider (isTrigger enabled)
- ☐ Rigidbody (Kinematic)
- ☐ Transform
- ☐ Animator Correct Answer: Collider (isTrigger enabled)

 **Will be reviewed**

10. A Scriptable Render Pipeline (SRP) in Unity is: 

- ☒ A system for writing custom rendering logic
- ☐ Used for animation blending
- ☐ A physics engine plugin
- ☐ An input manager Correct Answer: A system for writing custom rendering logic

 **Will be reviewed**

11. What does Time.deltaTime represent? 

- ☐ Time since game start
- ☒ Time between the current and previous frame
- ☐ Total simulation time
- ☐ Frame rate Correct Answer: Time between the current and previous frame

 **Will be reviewed**

12. In Unity's C# API, what does LateUpdate() commonly handle? 

- ☐ Physics

- ☒ Camera follow logic
- ☐ Input
- ☐ Initialization Correct Answer: Camera follow logic

 **Will be reviewed**

13. Which file format is not natively supported for 3D models in Unity? 

- ☐ FBX
- ☐ OBJ
- ☒ STL
- ☐ DAE Correct Answer: STL

 **Will be reviewed**

14. What's the function of the RenderTexture class? 

- ☐ Save a scene
- ☒ Store real-time rendered images
- ☐ Handle physics materials
- ☐ Manage scripts Correct Answer: Store real-time rendered images

 **Will be reviewed**

15. The main difference between Start() and Awake() is: 

- ☒ Awake() is called before Start()
- ☐ Start() is called for static variables
- ☐ Awake() runs every frame

☐ Awake() is slower Correct Answer: Awake() is called before Start()

 **Will be reviewed**

16. In VR, motion-to-photon latency refers to: 

- ☒ Delay between head movement and frame display
- ☐ Delay between controller input and haptic feedback
- ☐ Network delay
- ☐ Audio lag Correct Answer: Delay between head movement and frame display

 **Will be reviewed**

17. The Post-Processing Stack in Unity is used for: 

- ☐ Scene optimization
- ☒ Visual effects like bloom and color grading
- ☐ Physics simulation
- ☐ Shader compilation Correct Answer: Visual effects like bloom and color grading

 **Will be reviewed**

18. What's the purpose of a Normal Map? 

- ☐ Add color
- ☒ Simulate surface details without more geometry
- ☐ Control transparency
- ☐ Adjust UV scaling Correct Answer: Simulate surface details without more geometry

 **Will be reviewed**

19. Which Unity feature helps combine multiple materials into a single draw call? 

- ☒ Dynamic Batching
- ☐ LOD Group
- ☐ Reflection Probe
- ☐ Post Processing Correct Answer: Dynamic Batching

 **Will be reviewed**

20. In XR, positional tracking differs from rotational tracking because it: 

- ☐ Tracks only head rotation
- ☒ Tracks movement in 3D space
- ☐ Disables headset sensors
- ☐ Adjusts IPD automatically Correct Answer: Tracks movement in 3D space

 **Will be reviewed**

21. The Physics Raycast function can be used for: 

- ☒ Selecting objects using gaze
- ☐ Applying textures
- ☐ Creating particle trails
- ☐ Generating lighting data Correct Answer: Selecting objects using gaze

 **Will be reviewed**

22. In C#, which of the following destroys a GameObject? 

- ☐ Disable()
- ☐ Delete()
- ☒ Destroy(gameObject);
- ☐ Remove(gameObject); Correct Answer: Destroy(gameObject);

 **Will be reviewed**

23. GPU Instancing in Unity is used to: 

- ☒ Render multiple copies of a mesh efficiently
- ☐ Improve CPU-based AI
- ☐ Animate UI
- ☐ Handle post-processing Correct Answer: Render multiple copies of a mesh efficiently

 **Will be reviewed**

24. In Unity XR, hand-tracking input is supported through: 

- ☒ XR Plugin Management
- ☐ URP
- ☐ Render Texture
- ☐ Asset Bundles Correct Answer: XR Plugin Management

 **Will be reviewed**

25. What does DontDestroyOnLoad() do? 

- ☒ Keeps a GameObject across scene loads



- ☐ Saves data permanently
- ☐ Disables object pooling
- ☐ Freezes transform updates Correct Answer: Keeps a GameObject across scene loads

 **Will be reviewed**

26. The LOD Group component in Unity helps to: 

- ☐ Control physics layers
- ☒ Switch models based on camera distance
- ☐ Manage audio levels
- ☐ Handle animations Correct Answer: Switch models based on camera distance

 **Will be reviewed**

27. Foveated Rendering in XR reduces GPU load by: 

- ☐ Rendering per-eye depth buffers
- ☒ Rendering at high detail only where the user looks
- ☐ Lowering refresh rate
- ☐ Limiting color channels Correct Answer: Rendering at high detail only where the user looks

 **Will be reviewed**

28. The Profiler window in Unity is used to: 

- ☐ Create particle effects
- ☒ Measure performance and bottlenecks
- ☐ Edit materials

☐ Bake lighting Correct Answer: Measure performance and bottlenecks

 **Will be reviewed**

29. Shader Graph in Unity allows developers to: 

- ☐ Write C# shaders
- ☒ Visually build shaders without code
- ☐ Animate textures
- ☐ Manage post-processing Correct Answer: Visually build shaders without code

 **Will be reviewed**

30. In XR development, IPD (Interpupillary Distance) calibration is essential for: 

- ☐ Audio clarity
- ☒ Correct stereo image alignment
- ☐ Texture scaling
- ☐ Reducing draw calls Correct Answer: Correct stereo image alignment



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